



CSE 495: IT Project Management and Entrepreneurship

Section :04

Team Name : Sustainability Syndicate

Project on “Design and Development of a Sustainable Second-Hand Digital Marketplace with Trust and Authenticity Verification (Swapaholic)”

Submitted to

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Sustainable Second-Hand Digital Marketplace with Trust and Authenticity Verification (Swapaholic)

1. SCENERIO

Swapaholic is a web-based, AI-assisted second-hand marketplace designed to resolve trust, pricing, and safety issues commonly found in informal resale platforms in Bangladesh. The system operates through a centralized workflow where product listing, verification, pricing, transaction, and delivery are all managed within a single digital platform. Users interact with the system through a responsive web interface that functions seamlessly on both desktop and mobile browsers, eliminating the need for a native mobile application while maintaining accessibility for smartphone users.

In a typical scenario, a seller first accesses the Swapaholic website and creates an account by completing a structured registration process. To ensure platform integrity and reduce fraudulent activity, the seller is required to verify their identity through phone number OTP confirmation, email verification, submission of national identification information, and address details. Each completed verification step contributes to the seller's profile points, which represent credibility within the system and determine access to advanced features such as Cash-on-Delivery eligibility and reduced service fees. This verification process establishes an initial trust baseline before any product is listed on the platform.

After completing verification, the seller proceeds to list a second-hand product, such as an electronic device, by uploading product images and entering structured information including brand, model, condition, age, warranty status, and location. Once the information is submitted, the system automatically activates its AI modules. The uploaded images are analysed using a vision-based service to assess visual quality and consistency, while the textual and structured attributes are evaluated to generate an AI-based trust score. At the same time, the price prediction model processes historical marketplace data and product attributes to calculate a fair market price. Based on these evaluations, the system generates a recommended price range and an AI-assisted product description, which the seller can review before publishing the listing.

From the buyer's perspective, the platform allows users to browse verified listings through search and filtering options based on category, brand, price range, and location. Each product listing displays transparency-focused information, including the AI-generated trust score, verification status, recommended fair price, and seller credibility indicators. After reviewing a product, a buyer may either place a direct purchase request or submit a bid within an allowed negotiation range. If multiple bids are placed, the system automatically prioritizes the top offers and notifies the seller, allowing them to make an informed decision without manual negotiation overhead.

Once the seller confirms a buyer's request or accepts a bid, the transaction enters the quality control phase. A Quality Controller assigned by the platform visits the seller's location to physically inspect the product and verify its condition and authenticity. This step ensures that the product matches the description and images provided during listing. If the product passes inspection, it is temporarily stored in the platform's inventory for a limited period and marked as verified. If the product fails the inspection, the transaction is cancelled, the buyer is notified, and the seller's trust metrics may be adjusted accordingly.

After successful verification, the delivery process is initiated. A delivery partner is assigned, and the system tracks the product until it reaches the buyer. Depending on the buyer's eligibility and selected payment method, the transaction is completed either through online payment or Cash-on-Delivery. Upon confirmation of successful delivery, the system releases the payment to the seller within a predefined

timeframe, updates the transaction status, and removes the product from public listings. Both the buyer and seller receive profile point rewards, reinforcing responsible behaviour and trust within the platform.

Throughout the entire process, administrators monitor platform activities through a management interface that allows oversight of users, listings, quality control operations, and delivery workflows. This administrative supervision ensures compliance with platform rules, detects suspicious behaviour, and maintains system reliability. Overall, this scenario demonstrates how Swapaholic functions as a structured, AI-driven web marketplace that enhances trust, transparency, and safety in second-hand transactions while supporting sustainable consumption practices.

2. Project Charter

PROJECT CHARTER					
Project Title	Design and Development of a Sustainable Second-Hand Digital Marketplace with Trust and Authenticity Verification (Swapaholic)			Project Manager	Mehedi Hasan
Project Start Date	12-08-2025	Project End Date	30-04-2026	Project Sponsor	Tanni Mitra
Business Need					
1. The second-hand market suffers from low consumer trust, frequent fraud, and fragmented transaction flows that discourage buyers and sellers from participating at scale. 2. Businesses require a secure, verified marketplace that enforces authenticity checks, reliable payment handling, and delivery-based verification to reduce risk and improve transaction completion. 3. Investing in Swapaholic’s integrated verification, rating, and admin systems will unlock resale market growth, promote sustainable reuse, and create a scalable, revenue-generating platform while mitigating regulatory and operational risks.					
Purpose of the Project					
Develop Swapaholic, a mobile-first second-hand marketplace that restores buyer and seller trust through delivery-based product verification, secure payments, ratings, and dispute resolution. Deliver a validated MVP with admin and delivery management modules to enable scalable, fraud-resistant transactions and support future features like blockchain product IDs and AI-driven pricing.					
Project High Level Risk & Issues					
Risk	Probability	Impact	Mitigation		
Low user trust in early stages	Medium	Slow adoption and low transaction volume	Implement strict verification, rating system, and clear refund/dispute policies		
Fraudulent or misrepresented products	High	Financial loss and reputation damage	Admin review, delivery-time product verification, and seller reputation scoring		
Delivery and logistics delays	Medium	Customer dissatisfaction and order cancellations	Partner with reliable delivery vendors and add real-time tracking		
Payment or transaction failures	Low–Medium	Loss of user confidence and disputes	Use secure payment gateways, escrow system, and automated transaction logs		
Technical scalability issues	Medium	System downtime or poor performance	Use scalable cloud infrastructure and conduct load testing		
Slow manual admin verification process	Medium	Operational bottlenecks and delayed listings	Introduce automated checks and gradually integrate AI-based validation		
Key Deliverables				Milestones of the Project	
Deliverable	Description	Success Criteria	Milestone	Target Date	
Mobile Marketplace Application	User-facing app for browsing, listing, buying, and selling second-hand products	Users can register, list items, place orders, and complete transactions without critical errors	Project Initiation & Charter Approval	12-08-2025	
Admin Dashboard	Web-based panel for product approval, pricing checks, dispute handling, and user management	Admin can review listings, manage disputes, and monitor system activity efficiently	Requirements Analysis Completed	15-09-2025	
Delivery & Verification Module	System to manage delivery partners and enable in-person product verification before payment release	Successful completion of delivery-based verification and reduced dispute rates	System Design & Architecture Finalized	15-10-2025	
Secure Payment & Escrow System	Payment handling with escrow logic to release funds only after buyer confirmation	Zero critical payment failures and secure transaction records	Core Development Phase Completed	15-01-2026	
Rating & Review System	Mechanism for buyers and sellers to rate each other after transactions	Visible reputation scores and increased user trust indicators	Integration & System Testing Completed	15-03-2026	
MVP Deployment & Testing Report	Deployed minimum viable product with usability, performance, and security testing results	System passes core test cases and is ready for pilot launch	User Acceptance Testing (UAT)	10-04-2026	
			Final Deployment & Documentation Submission	30-04-2026	

Project Resources			Budget Summary	
Resource Type	Role / Vendor	Responsibility	Category	Estimated Cost (BDT)
Staff	Project Manager	Overall planning, coordination, timeline and budget control	Development (App, Backend, Admin Panel)	9,00,000
Staff	Frontend Developer (React / Mobile)	Develop user interface and mobile-first marketplace features	Infrastructure & Hosting (Cloud, Database, Security)	2,00,000
Staff	Backend Developer (Node.js / Express / MongoDB)	API development, database design, authentication (JWT), business logic	Delivery & Operations Setup	1,50,000
Staff	UI/UX Designer	User flow design, wireframing, usability optimization	Testing & Quality Assurance	75,000
Staff	QA Engineer	Testing (functional, security, performance), bug tracking	Marketing & User Acquisition (Pilot Phase)	75,000
Staff	Admin/Operations Executive	Product verification, dispute handling, platform monitoring	Contingency (10%)	1,00,000
Vendors	Cloud Hosting Provider (e.g., AWS/DigitalOcean)	Server infrastructure, database hosting, scalability	Total	15,00,000
Vendors	Payment Gateway Provider	Secure payment processing and escrow handling		
Vendors	Delivery/Logistics Partner	Product pickup, delivery, and in-person verification support		
Vendors	SMS/Email Service Provider	Notifications, OTP verification, transaction alerts		
Assumptions and Constraints				
Assumptions		Constraints		
Users will have access to smartphones with stable internet connectivity to use the mobile-first marketplace.		Total project budget is limited to 15,00,000 BDT.		
Payment gateway and delivery partners will be available and operational throughout the project lifecycle.		Project timeline is fixed from 12-08-2025 to 30-04-2026.		
The selected technology stack (React, Node.js, MongoDB, JWT) will meet performance and scalability requirements.		The system must be delivered as a functional MVP within the capstone project schedule.		
Admin verification processes will be sufficient to maintain product authenticity and user trust during the MVP stage.		The project must use the predefined technology stack and development approach.		
The project team will remain available as planned, with no major changes in key technical roles.		Manual product verification and in-house delivery model may limit operational scalability in the early phase.		
Initial target users will be willing to adopt a verified second-hand marketplace model.		Compliance with local regulations, payment policies, and data protection rules must be maintained.		
Project Team			Approval / Review Committee	
Project Manager	Mehedi Hasan		Sponsor	Tanni Mitra
Project Manager	Tonmoy Das		Business Division Head	Kafi
Team Member	Jarin, Sadia , Maisha		Finance Manager	Sadia Reza

3. Stakeholder Register

3.1 Stakeholder Identification Information

ID	Name / Role	Organization	Contact	Position / Authority
S1	Project Supervisor (Sponsor)	East West University	supervisor@ewubd.edu	Provides guidance, reviews progress, and approves key deliverables
	Project Manager / Team Lead	Swapaholic Development Team	teamlead@swapaholic.com	Leads execution, manages tasks, and ensures timely completion
	AI Engineer	Swapaholic Development Team	ai.dev@swapaholic.com	Develops and maintains AI models for trust rating and price prediction
	Backend Developer	Swapaholic Development Team	backend@swapaholic.com	Responsible for API development, database, and system integration
	Frontend Developer / UI Designer	Swapaholic Development Team	frontend@swapaholic.com	Designs and develops the user interface and user experience
	Quality Controller (QC Agent)	Swapaholic Operations	qc@swapaholic.com	Verifies product authenticity and ensures inspection workflow
	End Users (Buyers & Sellers)	Swapaholic Platform	users@swapaholic.com	Use the platform, provide feedback, and influence feature priorities
	Logistics Partner	Steadfast Courier / Delivery Partner	logistics@steadfast.com	Manages product pickup and delivery operations
	Payment Gateway Provider	Stripe / Local Payment Partner	payment@stripe.com	Handles secure transactions and payment settlements

3.2 Assessment Information

ID	Main Requirements	Expectations	Influence Level	Interest Level	Key Phase Interest
S1	Proper documentation + grading checklist met	Everything required by template is present	Very High	Medium	Initiation, Closing
S2	Timeline control + integration readiness	All modules merge without breaking	High	Very High	Planning, Execution
S3	Smart price suggestion / recommendations	Model improves user decision-making	Medium	High	Planning, Execution
S4	Fast APIs, secure user data, scalable DB	o crashes, no data loss, clean deployment	High	High	Planning, Execution
S5	Minimal clicks to buy/sell, clean UI	Even first-time users can complete tasks	Medium	Very High	Planning, Execution
S6	Reports/flagging + moderation tools	Bad listings removed quickly	Medium	High	Execution
S7	Smooth buying/selling, trust signals, order updates	I can buy safely and sell quickly	Medium	Very High	Execution, Maintenance
S8	Delivery info accuracy + tracking	No confusion on pickup and address	Medium	Medium	Execution
S9	Secure transactions + refund process	Payments always succeed or fail gracefully	High	Medium	Execution

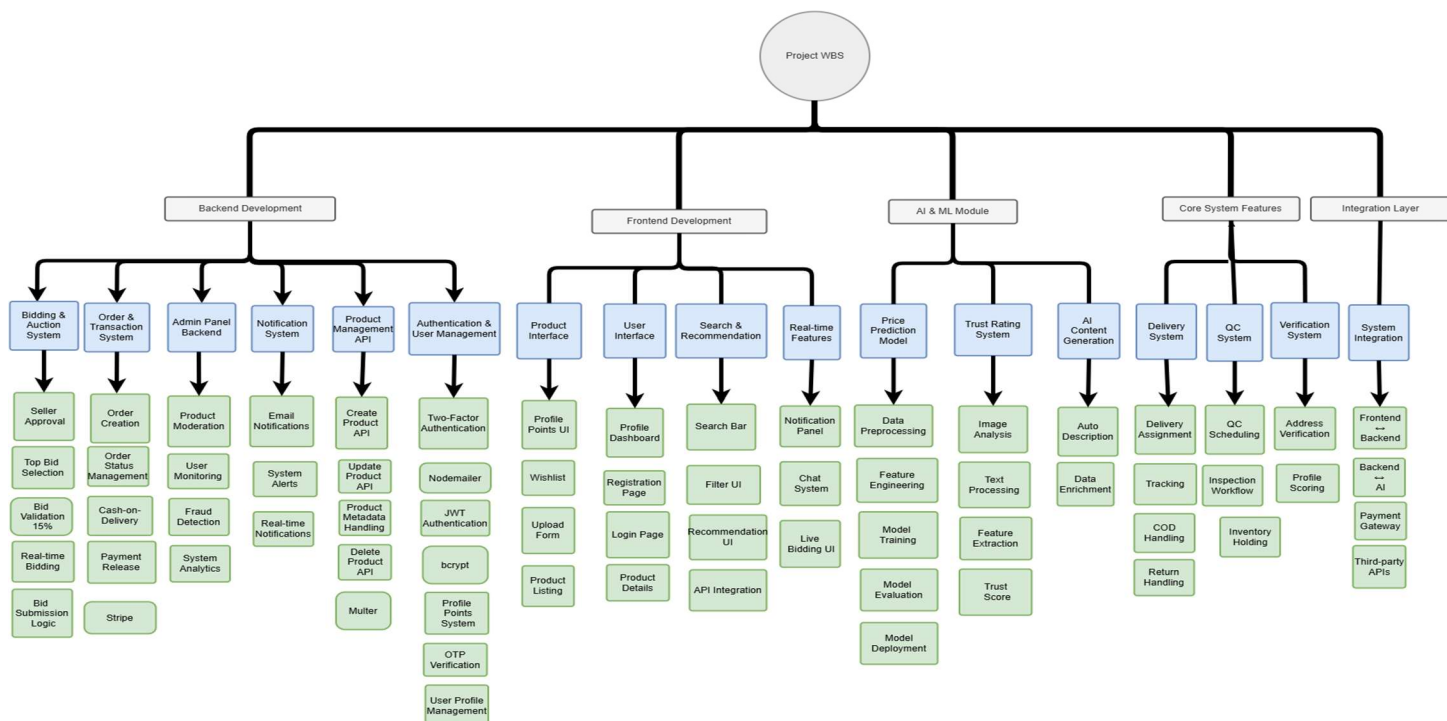
3.3 Stakeholder Classification

ID	Type	Attitude	Category
S1	Internal	Positive	Instructor / Evaluator
S2	Internal	Positive	Project Coordination Lead
S3	Internal	Neutral	Data & Pricing Analyst
S4	Internal	Positive	Application Development Team
S5	Internal	Positive	User Experience Design Team
S6	Internal	Neutral	Verification & Compliance Team
S7	External	Neutral	Marketplace Users
S8	External	Neutral	Delivery Service Partner
S9	External	Cautious	Digital Payment Service Provider

3.4 Stakeholder Management Strategy

Stakeholder Group	Input From Other Sections	Your Management Strategy (HOW)
Instructor / Evaluator (S1)	High influence, documentation-focused	Formal progress reports and milestone reviews will be conducted to ensure academic requirements are met.
Technical & Integration Team (S2, S4)	High system dependency and risk	Agile sprint meetings, integration checkpoints, and technical reviews will be used to manage coordination.
AI/ML Contributors (S3)	Decision-support expectation	Periodic evaluation of model output and feedback-based refinement will be applied.
UI/UX Designers (S5)	High user experience impact	Usability testing and design validation sessions will guide interface decisions.
Admin & Moderation Team (S6)	Platform control responsibility	Monitoring dashboards and escalation rules will be used for quick intervention.
End Users – Buyers & Sellers (S7)	Trust and usability sensitive	Transparent communication, verification indicators, and order tracking will be maintained.
Delivery Experts (S8)	Operational dependency	Clear task instructions, verification checklists, and tracking tools will be enforced.
Payment Partners (S9)	Financial risk exposure	Secure API integration, transaction logging, and refund workflows will be monitored.

4. WBS Tree Diagram



5. Product Backlog

Module	ID	User Story	Initial Estimation	Sprint	Sprint 1 Done	Sprint 1 Remain	Sprint 2 Done	Sprint 2 Remain	Sprint 3 Done	Sprint 3 Remain	Sprint 4 Done	Sprint 4 Remain	Sprint 5 Done	Sprint 5 Remain
1. User Management	1.1	As a user, I want to create an account and login securely	10	1	10	0	0	0	0	0	0	0	0	0
1. User Management	1.2	As a user, I want to verify my account using OTP/email	10	1	10	0	0	0	0	0	0	0	0	0
1. User Management	1.3	As a user, I want to update my profile information	5	1	5	0	0	0	0	0	0	0	0	0
1. User Management	1.4	As a user, I want to view my account activity and history	10	1	10	0	0	0	0	0	0	0	0	0
1. User Management	1.5	As a user, I want to delete my account	10	1	10	0	0	0	0	0	0	0	0	0
2. Product Management	2.1	As a seller, I want to upload product details	15	1	15	0	0	0	0	0	0	0	0	0
2. Product Management	2.2	As a seller, I want to upload product images	10	1	10	0	0	0	0	0	0	0	0	0
2. Product Management	2.3	As a seller, I want to edit a product listing	10	1	5	5	5	0	0	0	0	0	0	0
2. Product Management	2.4	As a seller, I want to delete a product listing	10	2	0	0	10	0	0	0	0	0	0	0
3. Shopping Cart and Checkout	3.1	As a buyer, I want to add items to my cart	10	2	0	0	10	0	0	0	0	0	0	0
3. Shopping Cart and Checkout	3.2	As a buyer, I want to update the quantity of items in my cart	5	2	0	0	5	0	0	0	0	0	0	0
3. Shopping Cart and Checkout	3.3	As a buyer, I want to remove items from my cart	5	2	0	0	5	0	0	0	0	0	0	0
3. Shopping Cart and Checkout	3.4	As a buyer, I want to review my cart before checkout	10	2	0	0	10	0	0	0	0	0	0	0
3. Shopping Cart and Checkout	3.5	As a buyer, I want to enter shipping/billing info and place the	15	2	0	0	15	0	0	0	0	0	0	0
4. AI Module	4.1	As a seller, I want AI-based price prediction	20	2	0	0	20	0	0	0	0	0	0	0
4. AI Module	4.2	As a buyer, I want AI-based trust rating for products	15	3	0	0	0	0	15	0	0	0	0	0
4. AI Module	4.3	As a system, I want product image analysis	15	3	0	0	0	0	15	0	0	0	0	0
5. Search & Recommendation	5.1	As a user, I want to search products	10	3	0	0	0	0	10	0	0	0	0	0
5. Search & Recommendation	5.2	As a user, I want to filter products	10	3	0	0	0	0	10	0	0	0	0	0
5. Search & Recommendation	5.3	As a user, I want personalized recommendations	15	3	0	0	0	0	15	0	0	0	0	0
6. Bidding System	6.1	As a buyer, I want to place bids on products	15	3	0	0	0	0	15	0	0	0	0	0
6. Bidding System	6.2	As a seller, I want to accept/reject bids	10	4	0	0	0	0	0	0	10	0	0	0
6. Bidding System	6.3	As a system, I want to select the top 3 bids	10	4	0	0	0	0	0	0	10	0	0	0
7. QC System	7.1	As a system, I want to schedule QC inspection	10	4	0	0	0	0	0	0	10	0	0	0
7. QC System	7.2	As a QC agent, I want to verify product authenticity	15	4	0	0	0	0	0	0	15	0	0	0
7. QC System	7.3	As a system, I want to update QC status	10	4	0	0	0	0	0	0	10	0	0	0
8. Delivery System	8.1	As a system, I want to assign delivery personnel	10	4	0	0	0	0	0	0	10	0	0	0
8. Delivery System	8.2	As a user, I want to track delivery status	10	4	0	0	0	0	0	0	10	0	0	0
8. Delivery System	8.3	As a system, I want to complete delivery workflow	15	5	0	0	0	0	0	0	5	10	10	0
9. Payment System	9.1	As a buyer, I want to initiate COD/online payment	15	5	0	0	0	0	0	0	0	0	15	0
9. Payment System	9.2	As a system, I want to release payment to seller	10	5	0	0	0	0	0	0	0	0	10	0
9. Payment System	9.3	As a system, I want to handle payment status/failure	10	5	0	0	0	0	0	0	0	0	10	0
10. Admin System	10.1	As an admin, I want to manage users	10	5	0	0	0	0	0	0	0	0	10	0
10. Admin System	10.2	As an admin, I want to monitor fraud activities	15	5	0	0	0	0	0	0	0	0	15	0
10. Admin System	10.3	As an admin, I want to manage products and listings	15	5	0	0	0	0	0	0	0	0	15	0
Sprint Totals					75	5	80	0	80	0	80	10	85	0

6. Sprint wise Calculation

6.1 Sprint 1

Sprint 1				
Person	Level	Weight	Allocation %	Allocation Cost
A	Scrum Master	1.5	100.00%	1.5
B	Junior Developer	0.8	50.00%	0.4
C	Senior Developer	1	50.00%	0.5
D	Senior Developer	1	50.00%	0.5
E	Junior Developer	0.8	50.00%	0.4
F	Junior Developer	0.8	50.00%	0.4
G	Junior Developer	0.8	50.00%	0.4
H	Senior Developer	1	50.00%	0.5
			450.00%	4.6
		Cost in man weeks		4.6
		Cost in man days		23
		Actual Cost (AC)		20.00%
		Earned Value (EV)		18.75%
		Cumulative Earned Value		18.75%
		Cost Variance (CV)		-1.25%
		Planned Value (PV)		20.00%
		Cumulative Planned Value		20.00%
		Schedule Variance (SV)		-1.25%
		Cumulative Actual Cost		20.00%

6.2 Sprint 2

Sprint 2				
Person	Level	Weight	Allocation %	Allocation Cost
A	Scrum Master	1.5	100.00%	1.5
B	Junior Developer	0.8	50.00%	0.4
C	Senior Developer	1	50.00%	0.5
D	Senior Developer	1	50.00%	0.5
E	Junior Developer	0.8	50.00%	0.4
F	Junior Developer	0.8	50.00%	0.4
G	Junior Developer	0.8	50.00%	0.4
H	Senior Developer	1	50.00%	0.5
			450.00%	4.6
		Cost in man weeks		4.6
		Cost in man days		23
		Actual Cost (AC)		20.00%
		Earned Value (EV)		20.00%
		Cumulative Earned Value		38.75%
		Cost Variance (CV)		0.00%
		Planned Value (PV)		18.75%
		Cumulative Planned Value		38.75%
		Schedule Variance (SV)		1.25%
		Cumulative Actual Cost		40.00%

6.3 Sprint 3

Sprint 3				
Person	Level	Weight	Allocation %	Allocation Cost
A	Scrum Master	1.5	100.00%	1.5
B	Junior Developer	0.8	50.00%	0.4
C	Senior Developer	1	50.00%	0.5
D	Senior Developer	1	50.00%	0.5
E	Junior Developer	0.8	50.00%	0.4
F	Junior Developer	0.8	50.00%	0.4
G	Junior Developer	0.8	50.00%	0.4
H	Senior Developer	1	50.00%	0.5
			450.00%	4.6
		Cost in man weeks		4.6
		Cost in man days		23
		Actual Cost (AC)		20.00%
		Earned Value (EV)		20.00%
		Cumulative Earned Value		58.75%
		Cost Variance (CV)		0.00%
		Planned Value (PV)		20.00%
		Cumulative Planned Value		58.75%
		Schedule Variance (SV)		0.00%
		Cumulative Actual Cost		60.00%

6.4 Sprint4

Sprint 4				
Person	Level	Weight	Allocation %	Allocation Cost
A	Scrum Master	1.5	100.00%	1.5
B	Junior Developer	0.8	50.00%	0.4
C	Senior Developer	1	50.00%	0.5
D	Senior Developer	1	50.00%	0.5
E	Junior Developer	0.8	50.00%	0.4
F	Junior Developer	0.8	50.00%	0.4
G	Junior Developer	0.8	50.00%	0.4
H	Senior Developer	1	50.00%	0.5
			450.00%	4.6
		Cost in man weeks		4.6
		Cost in man days		23
		Actual Cost (AC)		20.00%
		Earned Value (EV)		20.00%
		Cumulative Earned Value		78.75%
		Cost Variance (CV)		0.00%
		Planned Value (PV)		18.75%
		Cumulative Planned Value		77.50%
		Schedule Variance (SV)		1.25%
		Cumulative Actual Cost		80.00%

6.5 Sprint 5

Sprint 5				
Person	Level	Weight	Allocation %	Allocation Cost
A	Scrum Master	1.5	100.00%	1.5
B	Junior Developer	0.8	50.00%	0.4
C	Senior Developer	1	50.00%	0.5
D	Senior Developer	1	50.00%	0.5
E	Junior Developer	0.8	50.00%	0.4
F	Junior Developer	0.8	50.00%	0.4
G	Junior Developer	0.8	50.00%	0.4
H	Senior Developer	1	50.00%	0.5
			450.00%	4.6
			Cost in man weeks	4.6
			Cost in man days	23
			Actual Cost (AC)	20.00%
			Earned Value (EV)	21.25%
			Cumulative Earned Value	100.00%
			Cost Variance (CV)	1.25%
			Planned Value (PV)	22.50%
			Cumulative Planned Value	100.00%
			Schedule Variance (SV)	-1.25%
			Cumulative Actual Cost	100.00%